

Revised Formal Semantics of the Extended BPMN Language

1 Process Model

An extended BPMN process model is

$$\mathcal{P} = \langle Part, FE, S, \kappa_P \rangle,$$

where $Part$ is the finite set of participants,

$$FE = Event \uplus Activity \uplus Gateway, \quad Event = Start \uplus End,$$

and

$$S \subseteq FE \times FE$$

is the sequence-flow relation. Gateway types are

$$gtype : Gateway \rightarrow \{Exclusive, Inclusive, Parallel\}.$$

For $a \in FE$, define

$$In(a) = \{(x, a) \in S\}, \quad Out(a) = \{(a, y) \in S\}.$$

2 State-Dependent BPMN Specifications

Let $\mathcal{A}$ be the shared ACL model. The BPMN annotation map κ_P induces

$$pre_P : FE \setminus End \rightarrow BoolExpr(\mathcal{A}),$$

$$post_P : FE \setminus End \rightarrow BoolPostExpr(\mathcal{A}),$$

$$guard : S \rightarrow BoolExpr(\mathcal{A}).$$

An omitted precondition, postcondition, or guard denotes *true*. This explicitly includes the SequenceFlow guards required by the paper.

3 Control State

Introduce a distinguished entry symbol $\perp$ and

$$S^0 = \{(\perp, s) \mid s \in Start\}, \quad S_B = S \cup S^0.$$

A BPMN control state is a token marking

$$q \subseteq S_B.$$

The initial marking is

$$q_0 = S^0.$$

A marking is complete iff every remaining token is located immediately before an End Event:

$$Complete(q) \iff q \neq \emptyset \wedge \forall f \in q : target(f) \in End.$$

End Events are terminal observations and are not executed further in this core profile.

4 Enabledness and Token Consumption

For $a \in FE \setminus End$, let $C \subseteq In(a) \cup S^0$ be the set of tokens consumed by one execution. The predicate $Enabled(a, q, C)$ is defined as follows.

For Start Events and ordinary Activities:

$$Enabled(a, q, C) \iff C = \{f\} \wedge f \in q \wedge target(f) = a.$$

For an Exclusive gateway:

$$Enabled(a, q, C) \iff C = \{f\} \wedge f \in q \cap In(a).$$

For a Parallel gateway (join semantics):

$$Enabled(a, q, C) \iff C = In(a) \wedge In(a) \subseteq q.$$

For an Inclusive gateway, the exact BPMN non-local OR-join rule depends on upstream reachability. To avoid the ambiguity of choosing an arbitrary subset, define an abstract predicate

$$ORJoinEnabled(a, q, C),$$

which must implement the standard inclusive-join condition for the chosen BPMN profile, and set

$$Enabled(a, q, C) \iff ORJoinEnabled(a, q, C).$$

For a pure split gateway with one incoming flow, the above rules reduce to consuming that single incoming token.

5 Outgoing-Flow Selection

Given ACL post-state σ' , define the guard-enabled outgoing flows

$$GOut(a, \sigma') = \{s \in Out(a) \mid \sigma' \models guard(s)\}.$$

The produced set O must satisfy $Select(a, \sigma', O)$:

$$Select(a, \sigma', O) \iff \begin{cases} O = Out(a), & a \in Start \cup Activity, \\ O \subseteq GOut(a, \sigma') \wedge |O| = 1, & gtype(a) = Exclusive, \\ \emptyset \neq O \subseteq GOut(a, \sigma'), & gtype(a) = Inclusive, \\ O = GOut(a, \sigma') = Out(a), & gtype(a) = Parallel. \end{cases}$$

Thus XOR selects exactly one true guarded branch, OR selects a nonempty set of true guarded branches, and AND takes all branches. If the modelling profile supports default flows, the default-flow rule must be added explicitly rather than encoded as an unconstrained choice.

6 Coupled BPMN–ACL Execution Step

A FlowElement a executes from (σ, q) to (σ', q') , written

$$(\sigma, q) \xrightarrow{a}_{\mathcal{P}} (\sigma', q'),$$

iff there exist C, O such that

$$\begin{aligned} Enabled(a, q, C), \quad & \sigma \models pre_P(a), \\ (\sigma, \sigma') \models post_P(a), \quad & \sigma' \in \Sigma_A, \\ Select(a, \sigma', O), \quad & q' = (q \setminus C) \cup O. \end{aligned}$$

This relation replaces the previous set-valued *Pass/Next* “functions”, which were not functions when several subsets were possible.

7 System State and Scenario

Let μ be the iStar intentional marking. A unified system state is

$$s = \langle \sigma, q, \mu \rangle, \quad \sigma \in \Sigma_A.$$

Executing a gives

$$s_i = \langle \sigma_i, q_i, \mu_i \rangle \xrightarrow{a}_{\mathcal{S}} \langle \sigma_{i+1}, q_{i+1}, \mu_{i+1} \rangle = s_{i+1},$$

where $(\sigma_i, q_i) \xrightarrow{a}_{\mathcal{P}} (\sigma_{i+1}, q_{i+1})$ and μ_{i+1} is obtained by evaluating the iStar semantics over σ_{i+1} . A system scenario is

$$\pi = \langle s_0, a_1, s_1, \dots, a_n, s_n \rangle,$$

and is complete iff $Complete(q_n)$.

8 Conformance Classification

Let Π be the set of complete realizable scenarios. For each $\pi \in \Pi$, let μ_n^π be its final iStar marking and $RootGoal$ the set of iStar root goals. Define

$$Conformant(\pi) \iff \forall g \in RootGoal : \mu_n^\pi(g) = \top,$$

$$Risky(\pi) \iff \exists g \in RootGoal : \mu_n^\pi(g) = \perp.$$

Then

$$\Pi = \emptyset \implies Unrealizable,$$

$$Strong \iff \Pi \neq \emptyset \wedge \forall \pi \in \Pi : Conformant(\pi),$$

$$Weak \iff (\exists \pi \in \Pi : Conformant(\pi)) \wedge (\exists \pi \in \Pi : \neg Conformant(\pi)),$$

$$NonConformance \iff \Pi \neq \emptyset \wedge \forall \pi \in \Pi : \neg Conformant(\pi).$$

Risk is reported separately from the three-way conformance verdict.